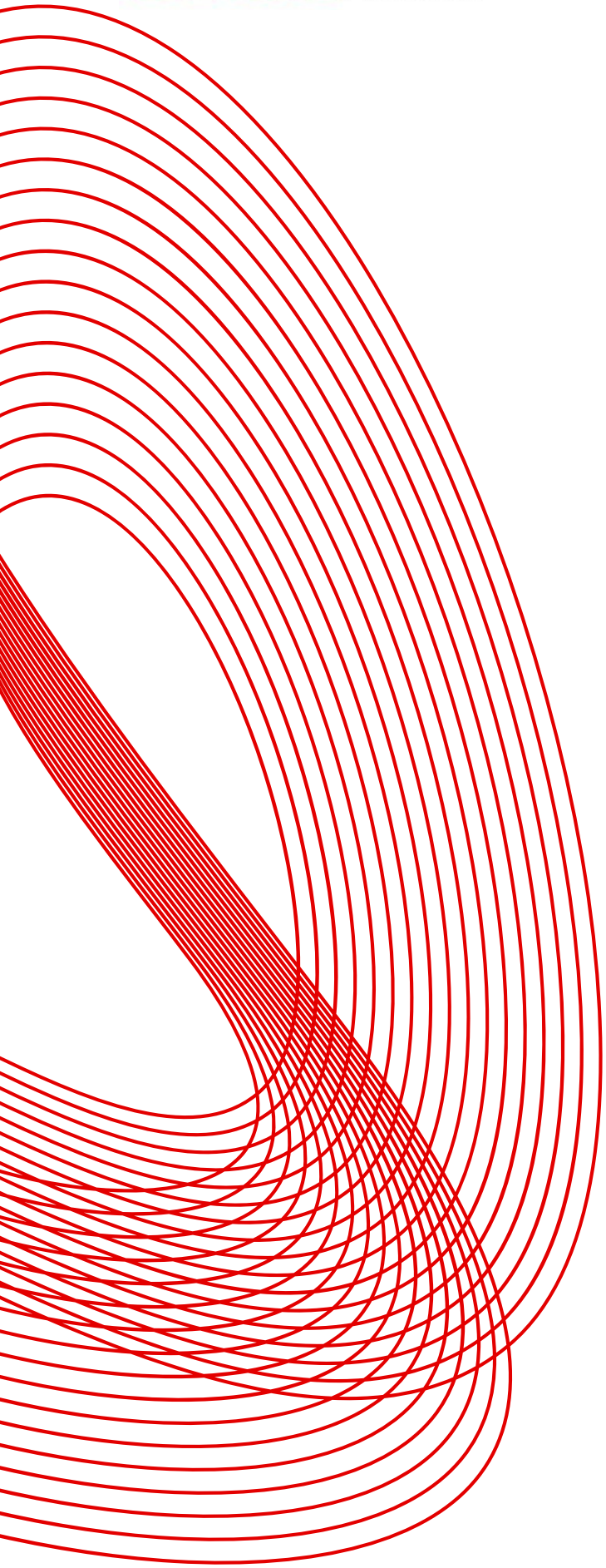
INSA INSTITUT NATIONAL DES SCIENCES APPLIQUÉES STRASBOURG

S9



S-IoT Project Report :

Drone swarm

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Specifications

Objective

The purpose of this project is to fly a swarm of UAVs (2 to 3 UAVs) and try to lift off a payload using a net.

Functions

- UWB precise location system
- PID Controller for drone stabilization
- UWB Communication for high frequency refreshing data
- Go back and forth keeping the drone formation (following a leader ?)

Deliverables

- Report with all progress, advancements and explanation of technical knowledge
- Functional UWB code
- Methodology and results of calibration of UWB sensors
- Bibliographical Studies
- Video of drone demo
- Functional PID Corrector
- Functional ESP32 code implementation using Multi-threading

Bibliographical Studies

0.1 UWB

0.1.1 Definitions

Explain what is UWB

0.1.2 Algorithm

Explain that we found and test different Algorithm

Two way ranging (TWR)

The Two Way Ranging (TWR) algorithm is the easiest and the fastest to implement in a UWB system. In order to explain this algorithm, we will use 2 modules called UWB_1 and UWB_2.

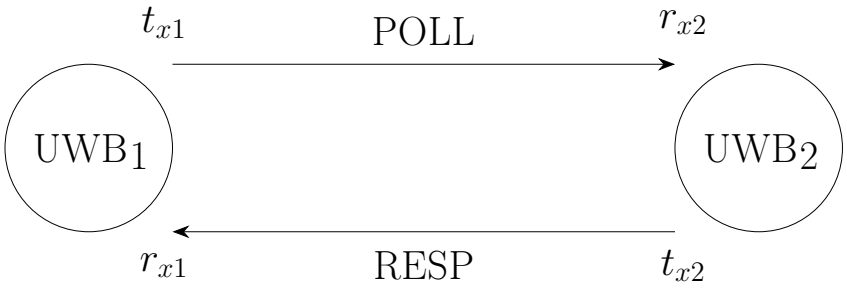


Figure 1: TWR explanation diagram

The algorithm works in 4 steps :

1. UWB_1 sends a **POLL** to UWB_2. The time of emission t_{x1} is memorized by UWB_1.
2. UWB_2 receives the **POLL** and memorizes the time of arrival r_{x2}.
3. UWB_2 processes the **POLL**, and precisely plans the time of emission t_{x2} for the **RESP**. It contains (t_{x2}, r_{x2}).
4. UWB_1 receives the **RESP** and memorizes the r_{x1}.

With all those parameters, we can deduce those variables :

- Total time : $t_{tot} = r_{x1} - t_{x1}$
- Processing time of UWB_2 : $t_{proc} = r_{x2} - t_{x2}$
- Travel time : $t_{travel} = t_{tot} - t_{proc}$

The signal travels back and forth so it did twice the distance (d), and travel speed of a radio signal is equivalent to light speed (c).
So, we can deduce this equation :

$$\boxed{\frac{t_{travel} \times c}{2} = \frac{(t_{tot} - t_{proc}) \times c}{2} = \frac{((r_{x1} - t_{x1}) - (r_{x2} - t_{x2})) \times c}{2}} \quad (1)$$

We can also determine the induced error of this algorithm.
After calculation (describe in calculation part at page 6), we obtain this equation:

$$\epsilon \approx \frac{t_{proc}(e_a - e_b)}{2} \tag{2}$$

As we can observe, the Time of processing is directly link to the error. The t_{proc} is around 3,000,000 nanoseconds so, it's inducing a high error.

Double Sided Two way ranging (DS-TWR)

The DS-TWR has for purpose to reduce the error introduce by the t_{proc}. This algorithm also use UWB_1 and UWB_2.

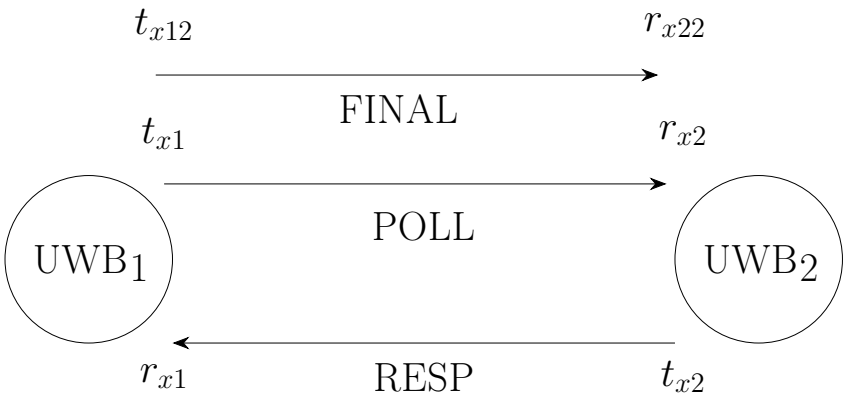


Figure 2: DS-TWR explanation diagram

0.2 Data treatment

0.2.1 Polynomial Filter

Caculations

TWR Error calculation :

Let's set:

- True time of flight : T_f
- True time of process : t_{proc}
- Clock drift errors : e_a and e_b

The measured times are:

$$R_a = (2T_f + t_{proc})(1 + e_a) \text{ and } t_{proc_b} = t_{proc}(1 + e_b) \tag{3}$$

The calculated time of flight is:

$$\hat{T}_f = \frac{R_a - t_{proc_b}}{2} \tag{4}$$

By replacing R_a and t_{proc_b}, we obtain :

$$\hat{T}_f = \frac{2T_f(1 + e_a) + t_{proc}(1 + e_a) - t_{proc}(1 + e_b)}{2} \tag{5}$$

By simplifying, we obtain :

$$\hat{T}_f = T_f(1 + e_a) + \frac{t_{proc}(e_a - e_b)}{2} \tag{6}$$

The error is $\hat{T}_f - T_f$, so:

$$Error = T_f \times e_a + \frac{t_{proc}(e_a - e_b)}{2} \tag{7}$$

The time of flight is about 33 nanoseconds, however D is around $3,000,000$.
So we can ignore $T_f \times e_a$ cause $T_f \ll t_{proc}$, and we obtain :

$$Error \approx \frac{t_{proc}(e_a - e_b)}{2} \tag{8}$$

DS-TWR Error calculation :